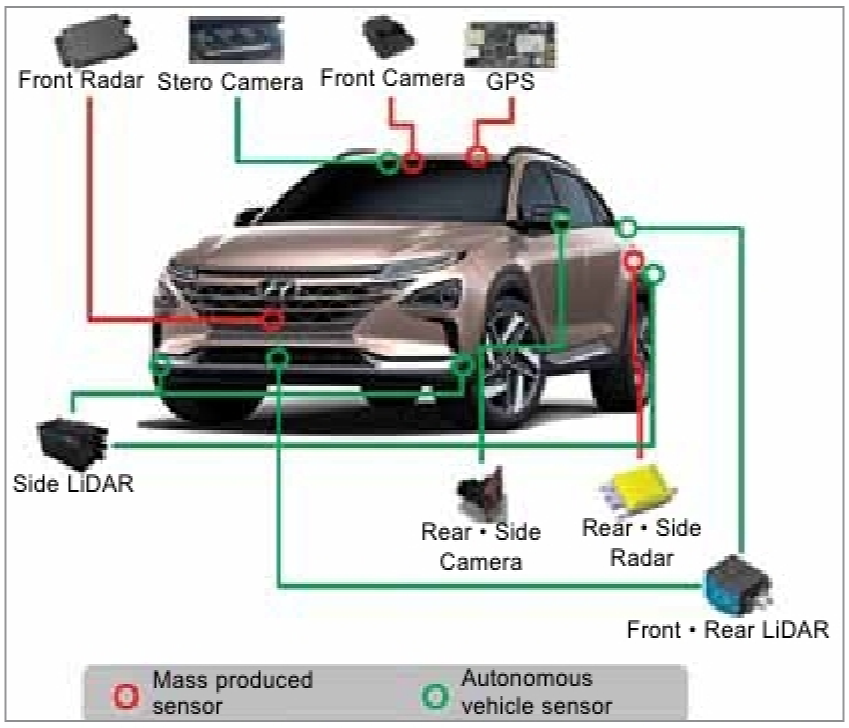


*Hyundai achieves world's first level-4 autonomous driving (Credit: https://automotiveleadnews.com)*

To support such software in a fully autonomous vehicle, the hardware requires an incredibly fast and powerful embedded system, such as the NVIDIA Drive AGX Pegasus platform, capable of 320 tera-operations per second (TOPS). Efforts are on to create a computer platform capable of solving complex algorithms in real time with minimal power consumption. In the coming years, some impressive technological breakthroughs using AI, ML, and data characterisation are expected.

## Autonomous micromobility

Micromobility refers to a range of small, lightweight devices operating at speeds typically below 25.6kmph and making trips under 15km. Micromobility devices include e-bikes, electric scooters, electric skateboards, shared bicycles, and electric pedal-assisted (pedelec) bicycles. This category has evolved to exclude devices with internal combustion engines and those with top speeds above 45kmph. This is one of the fastest-growing sectors of mobility in the world.

*One autonomous micromobility concept under development (Credit: https://www.yasuhideyokoi.com)*

Autonomous micromobility will help alleviate congestion by maximally packaging humans on roads. Uber is developing autonomous e-bikes and scooters that will drive themselves to charging stations. Autonomous micromobility will provide the ability to communicate road and traffic conditions to other vehicles using V2V technology along with the accelerated road-mapping capability to map roads better and faster.

## Technology for vehicles 'to see'

Capability 'to see' is the key to autonomous vehicle safety. Devices that are needed to make vehicles 'to see' the road around include cameras, light detection and ranging sensors (LiDAR), automotive radars, and ultrasonic sensors. Software is used for 'fusing' these different sources to localise the detected objects and predict where these will go next.

***Cameras.*** Cameras capture an image by recording the light reflected by the object's surface and provide clear images if the light intensity is sufficient. Cameras are continuously being made smaller, faster, higher-performing, and less expensive. Technology is available to decipher 3D images from the camera using a camera processing unit for detecting obstacles such as curbs and potholes. A new technology called DeepRay uses AI to reconstruct footages from damaged or obscured frames in real time to allow vehicles 'to see' during rain and snow.

***Ultrasonic sensors.*** An ultrasonic sensor measures an object's proximity by using a transducer to send and receive ultrasonic pulses relayed by the object. Ultrasonic sensing is usually used for short-distance applications at low speeds such as self-parking and blind-spot detection. The technology is also used in vehicle detection systems to automatically count the number of vehicles entering or leaving an area such as a parking lot and highway gates. In case of low light intensity, or when a medium interferes with the propagation of light, cameras provide poor quality images. This problem is solved by combining cameras with ultrasonic sensors that function well in conditions where light intensity is low, or the surroundings are dark. The advantages of ultrasonic sensors are their smaller size, relatively cheaper price, easier implementation, and lower power consumption.

***Automotive radars.*** Automotive radars are used to determine speed, direction, and range of objects in the vicinity of a vehicle by transmitting radio waves, which after hitting an object, get reflected and are received by radar. Frequency-modulated continuous-wave (FMCW) radars are simpler than pulse-doppler radars as the hardware required is considerably less complex, and the digital processing requirements are simpler. Automotive radars complement vision-based camera-sensing systems and are used in ADAS for collision avoidance and detection of pedestrians and cyclists.

***LiDAR sensors.*** LiDAR measures distance by illuminating the target with laser light and measuring the reflected light. Solid-state LiDAR technology provides sensors that can determine the velocity of the object as well as its distance from the radar. Efforts are on to make LiDAR sensors that can see as far as 200 metres with 2.5cm precision using only 100mW of power. Plus, these cost less than 500 dollars.

## Trends in electric vehicles (EVs) technologies

***Smaller and more efficient electric motors.*** Electric vehicles (EVs) are better performing, more efficient, luxurious, and above all more durable and less expensive. EVs can be made smaller with the same interior volume by reducing the electric motor size. Also, with reduced weight, a less powerful motor is required.

Brushless DC (BLDC) motors and AC induction motors are used in most of the electric vehicles considering